

Manasvi Kushwah

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Job Objective

Analytical and adaptable B.Tech undergraduate with strong communication, problem-solving, and teamwork skills. Experienced in projects and hackathons that involved presenting ideas and collaborating in fast-paced environments.

Education

VIT Bhopal University

2022 – 2026

B.Tech in Electronics and communication engineering

Specialization: AI and Cybernetics

Cgpa: 8.58/10

Technical Skills

Soft skills: Active listening, Communication, Team Management, Leadership

Languages: Python, C++, MATLAB, SQL

Projects

Low-Cost Spectrophotometer for Water Quality Analysis | Python, MATLAB, Signal Processing

Dec 2024 – May 2025

- Engineered a custom spectrophotometer using a webcam, diffraction grating, and photoresistor, focusing on hardware integration and optical calibration.
- Developed signal processing algorithms in MATLAB to analyze spectral data, identifying contaminants with a calibrated sensor accuracy of $\pm 5\%$ error.

ML Based Tuberculosis Detection using Cough Audio | Python, TensorFlow, Scikit-Learn

August 2025 – Dec 2025

- Built an end-to-end ML pipeline to classify TB from cough recordings .
- Extracted MFCC features and trained CNN models with transfer learning (Adam optimizer).
- Implemented SVM and ensemble techniques to improve robustness and performance.
- Evaluated using accuracy, precision, and recall metrics.

Sparse RBM for Blood Cancer Classification | Python, TensorFlow

Nov 2024 – Dec 2024

- Developed Sparse RBM for high-dimensional gene expression data.
- Achieved 82% classification accuracy for AML & ALL types

Certifications

Applied Machine Learning in Python - University of Michigan | Coursera

MATLAB Onramp, Signal Processing Onramp, & Computer Vision - MathWorks Academy

VLSI Design Training - Maven Silicon

Experience

Maven Silicon Internship in Embedded System Design | (Feb-May 2025)

- Demonstrated applied proficiency with Arduino UNO Rev3, Raspberry Pi 3, and Node MCU boards effectively integrated diverse sensors to gain knowledge of 3 Serial Communication protocols

Extracurriculars

- Smart India Hackathon (SIH) 2023 Participant : Shortlisted for the internal round for proposing an innovative robotics Solution (Hexabot)
- Participated in Jatayu Virtusa Ideathon as a Team Lead, focused on developing industry-oriented solutions, contributing to problem analysis, idea presentation, and collaborative strategy building.
- Collaborated in multiple academic projects requiring task coordination, active listening, and clear communication within cross-functional teams